

ZYLOVAA™

LOSARTAN POTASSIUM 50MG / 100MG
Film-Coated Tablet

COMPOSITION

Zylovaat Tablet 50mg: Each film-coated tablet contains 50mg Losartan Potassium.
Zylovaat Tablet 100mg: Each film-coated tablet contains 100mg Losartan Potassium.

PHARMACODYNAMICS

Losartan is an oral, specific angiotensin-II receptor (type AT₁) antagonist. Angiotensin II binds to the AT₁ receptor found in many tissues (e.g. vascular smooth muscle, adrenal gland, kidneys, and the heart) and elicits several important biological actions, including vasoconstriction and the release of aldosterone. Angiotensin II also stimulates smooth-muscle proliferation. Based on binding and pharmacological bioassays, it binds selectively to the AT₁ receptor.

During losartan administration, removal of angiotensin-II negative feedback on renin secretion leads to increased plasma renin activity. Increases in plasma renin activity lead to increases in angiotensin II in plasma. Even with these increases, antihypertensive activity and suppression of plasma aldosterone concentration are maintained, indicating effective angiotensin-II receptor blockade.

Losartan binds selectively to the AT₁ receptor and does not bind to or block other hormone receptors or ion channels important in cardiovascular regulation. Furthermore, losartan does not inhibit ACE (kininase II), the enzyme that degrades bradykinin. Consequently, effects not directly related to blocking the AT₁ receptor, such as the potentiation of bradykinin-mediated effects, the generation of oedema (losartan 1.7%, placebo 1.9%) or fatigue (losartan 3.8%, placebo 3.9%), are not associated with losartan.

Losartan has been shown to block responses to angiotensin I and angiotensin II without affecting responses to bradykinin. In contrast, ACE inhibitors have been shown to block responses to angiotensin I and enhance responses to bradykinin without altering the response to angiotensin II, thus providing a pharmacodynamic distinction between losartan and ACE inhibitors.

Plasma concentrations of losartan and its active metabolite, and the antihypertensive effect of losartan increase with increasing dose. Since losartan and its active metabolite are both angiotensin-II receptor antagonists, they both contribute to the antihypertensive effect.

In non-diabetic hypertensive patients with proteinuria, the administration of losartan potassium significantly reduces proteinuria, fractional excretion of albumin and IgG. Losartan maintains glomerular filtration rate and reduces filtration fraction. Generally, losartan causes a decrease in serum uric acid (usually <24 micromol) which was persistent in chronic therapy.

Losartan has no effect on autonomic reflexes and no sustained effect on plasma noradrenaline.

Losartan potassium administered in doses of up to 150 mg once daily did not cause clinically important changes in fasting triglycerides, total cholesterol or HDL cholesterol in patients with hypertension. The same doses of losartan had no effect on fasting glucose levels.

PHARMACOKINETICS

Absorption

Following oral administration, losartan is well absorbed and undergoes first-pass metabolism, forming an active carboxylic acid metabolite and other inactive metabolites. The systemic bioavailability of losartan tablets is approximately 33%. Mean peak concentrations of losartan and its active metabolite are reached in 1 hour and in 3-4 hours, respectively. There was no clinically significant effect on the plasma concentration profile of losartan when the drug was administered with a standardised meal.

Distribution

Both losartan and its active metabolite are ≥ 99% bound to plasma proteins, primarily albumin. The volume of distribution of losartan is 34 litres. Studies in rats indicate that losartan crosses the blood-brain barrier poorly, if at all.

Biotransformation

About 14% of an intravenously or orally-administered dose of losartan is converted to its active metabolite. In addition to the active metabolite, inactive metabolites are formed, including two major metabolites formed by hydroxylation of the butyl side chain and a minor metabolite, an N-2 tetrazole glucuronide.

Elimination

Plasma clearance of losartan and its active metabolite is about 600 ml/min and 50 ml/min, respectively. Renal clearance of losartan and its active metabolite is about 74 ml/min and 26 ml/min, respectively. When losartan is administered orally, about 4% of the dose is excreted unchanged in the urine, and about 6% of the dose is excreted in the urine as active metabolite. The pharmacokinetics of losartan and its active metabolite are linear with oral losartan potassium doses up to 200 mg.

Following oral administration, plasma concentrations of losartan and its active metabolite decline polyexponentially with a terminal half-life of about 2 hours and 6-9 hours, respectively. During once-daily dosing with 100 mg, neither losartan nor its active metabolite accumulates significantly in plasma.

Both biliary and urinary excretion contribute to the elimination of losartan and its metabolites.

Characteristics in patients

Following oral administration in patients with mild to moderate alcoholic cirrhosis of the liver, plasma concentrations of losartan and its active metabolite were, respectively, 5-fold and 1.7-fold greater than those seen in young male volunteers. Plasma concentrations of losartan are not altered in patients with creatinine clearance above 10 ml/min. Neither losartan nor the active metabolite can be removed by haemodialysis.

INDICATIONS

Hypertension

Zylovaat is indicated for the treatment of hypertension.

Hypertensive patients with left ventricular hypertrophy

Zylovaat is indicated to reduce the risk of stroke in patients with hypertension and left ventricular hypertrophy, but there is evidence that this benefit does not apply to Black patients.

Nephropathy in Type 2 Diabetic Patients

Indicated for the treatment of diabetic nephropathy with an elevated serum creatinine and proteinuria (urinary albumin to creatinine ratio 300 mg/g or more) in patients with type 2 diabetes and a history of hypertension. In this population, Zylovaat reduces the rate of progression of nephropathy as measured by the occurrence of doubling the serum creatinine or end stage renal disease (need for dialysis or renal transplantation) or death.

DOSAGE AND ADMINISTRATION

To be administered orally.

Zylovaat may be administered with or without food.

Zylovaat may be administered with other antihypertensive agents.

Hypertension

The starting and maintenance dose is 50 mg once daily for most patients. The maximal antihypertensive effect is attained 3-6 weeks after initiation of therapy. Some patients may receive an additional benefit by increasing the dose to 100 mg once daily.

For patients with intravascular volume depletion (e.g. those treated with high-dose diuretics), a starting dose of 25 mg once daily should be considered. (See Warnings and Precautions).

No initial dosage adjustment is necessary for elderly patients or for patients with renal impairment, including patients on dialysis. A lower dose should be considered for patients with a history of hepatic impairment.

Hypertensive patients with left ventricular hypertrophy

The usual starting dose is 50 mg of Zylovaat once daily. A low dose of hydrochlorothiazide may be added and/or the dose of Zylovaat should be increased to 100 mg once daily based on blood pressure.

Nephropathy in Type 2 Diabetic Patients

The usual starting dose is 50 mg once daily. The dose may be increased to 100 mg once daily based on blood pressure response. Zylovaat may be administered with other antihypertensive

agents (e.g., diuretics, calcium channel blockers, alpha- or beta-blockers, and centrally acting agents) as well as with insulin and other commonly used hypoglycaemic agents (e.g., sulfonylureas, glitazones and glucosidase inhibitors).

CONTRAINDICATIONS

Zylovaat is contraindicated in patients who are hypersensitive to any component of this product.

Concomitant use of losartan with aliskiren-containing products is contraindicated in patients with diabetes mellitus or renal impairment (GFR < 60 ml/min/1.73 m²).

WARNINGS AND PRECAUTIONS

Hypersensitivity: Angioedema. (See Side Effects).

Hypotension and electrolyte/fluid imbalance

In patients who are intravascularly volume depleted (e.g. those treated with high-dose diuretics), symptomatic hypotension may occur. These conditions should be corrected prior to administration of losartan, or a lower starting dose should be used. (See Dosage and Administration).

Electrolyte imbalances are common in patients with renal impairment, with or without diabetes, and should be addressed. In a clinical study conducted in type 2 diabetic patients with nephropathy, the incidence of hyperkalaemia was higher in the group treated with losartan as compared to the placebo group.

Liver function impairment

A lower dose should be considered for patients with a history of hepatic impairment. (See Dosage and Administration).

Renal function impairment

As a consequence of inhibiting the renin-angiotensin-aldosterone system, changes in renal function including renal failure have been reported in susceptible individuals; as with other drugs that affect the renin-angiotensin-aldosterone system, increases in blood urea and serum creatinine have also been reported in patients with bilateral renal artery stenosis or stenosis of the artery to a solitary kidney. These changes in renal function may be reversible upon discontinuation of therapy.

DRUG INTERACTIONS

No drug interactions have been identified with hydrochlorothiazide, digoxin, warfarin, cimetidine, ketoconazole, erythromycin and phenobarbital (phenobarbitone). Rifampicin and fluconazole have been reported to reduce levels of active metabolite. The clinical consequences of these interactions have not been evaluated.

As with other drugs that block angiotensin II or its effects, concomitant use of potassium-sparing diuretics (e.g. spironolactone, triamterene, amiloride), potassium supplements, or salt substitutes containing potassium may lead to increases in serum potassium.

As with other antihypertensive agents, the antihypertensive effect of losartan may be attenuated by the non-steroidal anti-inflammatory drug indomethacin.

As with other drugs which affect the excretion of sodium, lithium excretion may be reduced. Therefore, serum lithium levels should be monitored carefully if lithium salts are to be co-administered with angiotensin II receptor antagonists. In some patients with compromised renal function who are being treated with non-steroidal anti-inflammatory drugs, including selective cyclooxygenase-2 inhibitors, the co-administration of angiotensin II receptor antagonists may result in a further deterioration of renal function. These effects are usually reversible.

As with the ACE inhibitors, the use of losartan with an ACE inhibitor may increase the risk of hyperkalaemia, hypotension, and syncope, particularly in patients with atherosclerotic disease or heart failure, or in diabetics who have end-organ damage. Such combinations should be reserved for selected cases with close monitoring of renal function.

PREGNANCY AND LACTATION

Use during pregnancy

When used in pregnancy during the second and third trimesters, drugs that act directly on the renin-angiotensin-aldosterone system can cause injury and even death in the developing foetus. Losartan should not be used in pregnancy, and if pregnancy is detected losartan should be discontinued as soon as possible.

Use during lactation

It is not known whether losartan is excreted in human milk. Because of the potential for adverse effects on the nursing infant, a decision should be made whether to discontinue breast feeding or discontinue the drug, taking into account the importance of the drug to the mother.

SIDE EFFECTS

Side effects have usually been mild and transient in nature and have not required discontinuation of therapy. The most common drug-related side effects were dizziness, asthenia/fatigue, hypotension, hyperkalaemia and vertigo.

The following adverse reactions have been reported in post-marketing experience:

Hypersensitivity: Anaphylactic reactions, angioedema including swelling of the larynx and glottis causing airway obstruction and/or swelling of the face, lips, pharynx, and/or tongue have been reported rarely in patients treated with losartan; some of these patients previously experienced angioedema with other drugs including ACE inhibitors. Vasculitis, including Henoch-Schoenlein purpura, has been reported rarely.

Gastro-intestinal: Hepatitis (reported rarely), diarrhoea, liver function abnormalities.

Haematologic: Anaemia, thrombocytopenia (reported rarely).

Musculoskeletal: Myalgia, arthralgia.

Nervous system/Psychiatric: Migraine.

Respiratory: Cough.

Skin: Urticaria, pruritus, rash.

SYMPTOMS AND TREATMENT FOR OVERDOSAGE

The most likely manifestation of overdosage would be hypotension and tachycardia; bradycardia could occur from parasympathetic (vagal) stimulation. If symptomatic hypotension should occur, supportive treatment should be instituted. Neither losartan nor the active metabolite can be removed by haemodialysis.

STORAGE CONDITION

Store below 30°C. Protect from light.

SHELF-LIFE

The expiry date is indicated on the packaging.

PRODUCT DESCRIPTION

Zylovaat Tablet 50mg: White, oval, normal convex film-coated tablet with "Z" marking on one side and plain on reverse. Available as blister strips of 10's in packing of 30 tablets per box.

Zylovaat Tablet 100mg: White, round, normal convex film-coated tablet with "Z" marking on one side and plain on reverse. Available as blister strips of 10's in packing of 30 tablets per box.

KEEP OUT OF REACH OF CHILDREN / JAUHI DARI KANAK-KANAK

For further information, please consult your pharmacist or physician.

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Manufactured and Marketed by:

Xepa-Soul Pattinson (Malaysia) Sdn Bhd

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